

SQL Aggregate Functions & Opereratio

Exercise 2

Table: students

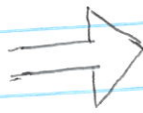
student-id	name	age	department
1	Alice	20	IT
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	Finance
5	Eve	22	HR

① SELECT DISTINCT department
FROM students;



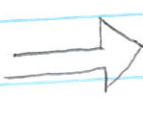
department
IT
HR
Finance

② SELECT department,
AVG(age) As avg-age
FROM students
GROUP BY department;



department	avg-age
IT	20.5
HR	22
Finance	23

③ SELECT department;
COUNT(*) As student-count
FROM students
GROUP BY department
HAVING COUNT(*) > 1;



department	student-count
IT	2
HR	2

④ SELECT student-id, name, age, department
 FROM students
 WHERE age BETWEEN 21 AND 23 ;



Student-id	name	age	department
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	Finance
5	Eve	22	HR

⑤ SELECT student-id, name,
 age, department
 FROM students
 WHERE age > 21 AND
 department = 'IT' OR
 department = 'HR' ;



Student-id	name	age	department
2	Bob	22	HR
5	Eve	22	HR

02 The Courses Table

Courses

Course-id	Course-name	department	credits
101	SQL Basics	IT	3
102	Python	IT	4
103	Data Science	IT	4
104	Excel	Finance	2
105	Statistics	HR	3

6) SELECT department,
SUM(credits) AS total_credits

FROM courses

GROUP BY department

HAVING SUM(credits) > 5;



department	total_credits
IT	11

7) SELECT course-id, course-name,
department, credit

FROM courses

WHERE credits < 4;



Course-id	Course-name	department	Credits
101	SQL Basics	IT	3
104	Excel	Finance	2
105	Statistics	HR	3

⑧ SELECT course-id, course-name, credits
FROM Courses
ORDER BY credits DESC
LIMIT 3;

⇓

Course-id	Course-name	Credits
101	SQL Basics	3
102	Python	4
103	Data Science	4

The ENROLLMENTS TABLE

⑨ enrollments

enrollment-id	student-id	course-id	grade
1	1	101	85
2	2	102	78
3	3	103	90
4	4	104	88
5	5	105	82

⑩ SELECT ~~max~~ MAX(grade) AS max-grade
FROM enrollments;



max-grade
90

SELECT MIN(grade) AS min-grade
FROM enrollments;



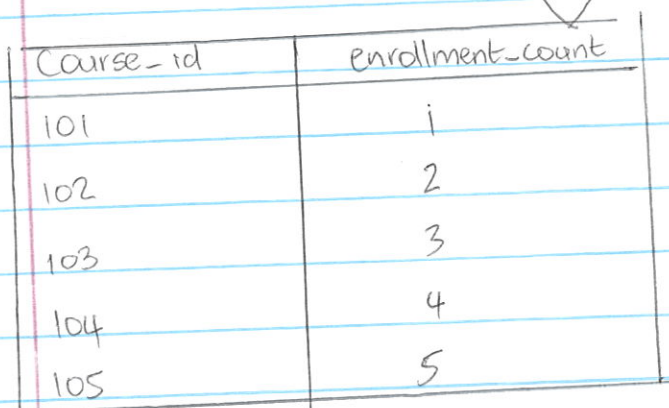
min-grade
78

SELECT AVG(grade) AS avg-grade
FROM enrollments;



avg-grade
84.6

10. SELECT course-id,
COUNT(*) AS enrollment_count
~~GROUP BY~~ course-id FROM enrollments
GROUP BY course-id;



Course-id	enrollment-count
101	1
102	2
103	3
104	4
105	5

④ a) THE SALARIES TABLE

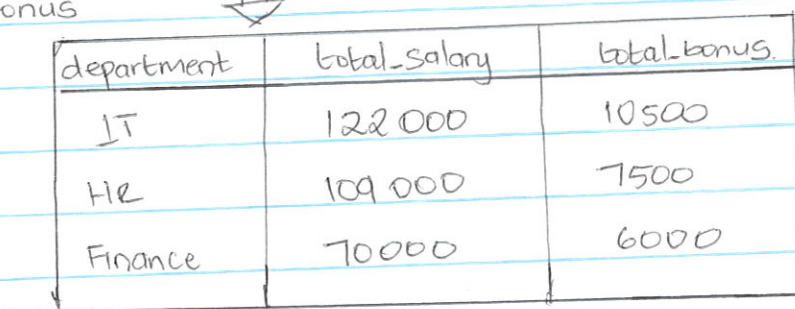
Salaries

employee-id	name	department	salary	bonus
1	Tom	IT	60000	9000
2	Jerry	HR	55000	4000
3	Spike	Finance	70000	6000
4	Tyke	IT	62000	5500
5	Butch	HR	54000	3500

⑪ SELECT department,
SUM(salary) AS total_salary
SUM(bonus) AS total_bonus

~~FROM~~ FROM salaries

GROUP BY department;



department	total_salary	total_bonus
IT	122000	10500
HR	109000	7500
Finance	70000	6000